



Indian Association for the Cultivation of Science
(Deemed to be University under the *de novo* category)

Quiz

Subject: Advanced Quantum Mechanics 1

Subject Code(s): (PHS 4102)

Full marks: 15

Time allotted: 1.0 hrs

1. A spin- $\frac{1}{2}$ particle of mass m with magnetic moment $\vec{\mu} = \mu_0 \vec{\sigma}$, which is moving in an infinite, symmetric potential well $V(x)$ of length $2L$, is placed in an external weak magnetic field $\vec{B}$ with:

$$V(x) = \begin{cases} 0, & -L \leq x \leq L, \\ \infty, & \text{otherwise.} \end{cases} \quad \vec{B} = \begin{cases} -B\hat{k}, & -L \leq x \leq 0, \\ -B\hat{i}, & 0 \leq x \leq L. \end{cases}$$

- (a) Write down the energy eigenvalues and wave functions of the unperturbed Hamiltonian. The wave function should include the spin part.
- (b) Using first order degenerate perturbation theory, calculate the energy of the ground state of this particle, considering $\hat{H}_p = -\vec{\mu} \cdot \vec{B}$

3+4

2. Use the WKB approximation to estimate the energy levels of a one-dimensional harmonic oscillator.

4

3. Use the trial function

$$\psi_0(x, \alpha) = \begin{cases} A(\alpha^2 - x^2), & |x| \leq \alpha, \\ 0, & |x| \geq \alpha, \end{cases}$$

to estimate the ground state energy of a one-dimensional harmonic oscillator by means of the variational method; α is an adjustable parameter and A is the normalization constant.

4



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Mid-Semester Examination

Subject: Advanced Quantum Mechanics 1

Subject Code(s): (PHS 4102)

Full marks: 25

Time allotted: 2.0 hrs

1. Consider the coherent state given by :

$$|\alpha\rangle = \exp\left(-\frac{1}{2}|\alpha|^2\right) \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle$$

where $|n\rangle$ is the eigenstate of the Hamiltonian for one dimensional harmonic oscillator and α is a complex number.

(a) Show that $|\alpha\rangle$ is an eigenstate of the annihilation operator a with eigenvalue α .

(b) Show the state $|\alpha\rangle$ is normalised.

The position and momentum operators in terms of creation and annihilation operators may be written as: $\hat{x} = \sqrt{\frac{\hbar}{2m\omega}}(a + a^\dagger)$ and $\hat{p} = i\sqrt{\frac{m\hbar\omega}{2}}(a^\dagger - a)$

(c) Show $\Delta x \Delta p = \frac{\hbar}{2}$ for the coherent state $|\alpha\rangle$

2+2+4

2. Calculate the CG coefficients needed to couple the two angular momenta $j_1 = \frac{1}{2}$ and $j_2 = \frac{1}{2}$ to the possible j values, and express the coupled states $|j_1 j_2 j m\rangle$ to the uncoupled basis $|j_1 j_2 m_1 m_2\rangle$.

6

3. A particle in a central potential has an orbital angular momentum $l = 2\hbar$ and a spin $s = \hbar$. Find the energy levels and degeneracies associated with a spin-orbit interaction term of the form $H_{soc} = A \vec{L} \cdot \vec{S}$, where A is a constant.

4

4. A particle of spin ($s=1/2$) is subject to the Hamiltonian $H = \alpha s_z + \beta s_x$, where α and β are constants.

(i) Calculate the energy levels of this system.

(ii) If at $t = 0$ the spin is in an eigenstate of s with $s_z = +\frac{\hbar}{2}$, calculate the expectation value of the spin at time t .

2+5



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End-Semester Examination

Subject: Advanced Quantum Mechanics 1

Subject Code(s): (PHS 4102)

Full marks: 50

Time allotted: 3.0 hrs

1. A two-state system evolves under the action of the Hamiltonian

$$H = E_0|A\rangle\langle A| + (E_0 + \Delta)|B\rangle\langle B|,$$

where $|A\rangle$ and $|B\rangle$ are its two orthonormal states. These states transform to one another under parity, i.e.

$$P|A\rangle = |B\rangle \quad \text{and} \quad P|B\rangle = |A\rangle.$$

- (a) Construct states with definite parity $P = +1$ and $P = -1$ at $t = 0$.
(b) If at time $t = 0$ the system is in a state of definite parity $P = 1$, find the earliest time t at which the probability of finding the system in a state of parity $P = -1$ is one.

1+3

2. Show that bound stationary states in one dimension are non-degenerate.

4

3. A spin- $\frac{1}{2}$ system is prepared in the initial state

$$|\varphi\rangle = \frac{\sqrt{3}}{2}|\uparrow\rangle + \frac{1}{2}|\downarrow\rangle$$

where $|\uparrow\rangle$ and $|\downarrow\rangle$ are eigenstates of $\hat{S}_z$ with eigenvalues $+\frac{\hbar}{2}$ and $-\frac{\hbar}{2}$ respectively.

- (a) What is the probability that the measurement of $\hat{S}_z$ yields a value $+\frac{\hbar}{2}$ or $-\frac{\hbar}{2}$?
(b) The measurement of $\hat{S}_z$ is followed by a measurement of $\hat{S}_x$ on the system. What is the probability that the measurement of $\hat{S}_x$ yields a value $+\frac{\hbar}{2}$?

2+4

4. Suppose you have two spin half particles, with their individual spin operators $\mathbf{S}_1$ and $\mathbf{S}_2$. Obtain the eigenstates of $\mathbf{S}^2$ and S_z , where $\mathbf{S} = \mathbf{S}_1 + \mathbf{S}_2$ using CG-coefficient.

5

- 5 (a) Directly compute the matrix elements:

$$\langle j = 1, m' | J_q | j = 1, m \rangle$$

where $J_q (q = -1, 0, 1)$ denotes the q -th spherical component of the angular momentum operator.

- (b) Verify that these matrix elements satisfy Wigner-Eckart theorem, and deduce the corresponding reduced matrix element from them.

(You may use the following CG coefficient: $\langle j, 1; m, 0 | j, m \rangle = \frac{m}{\sqrt{j(j+1)}}$)

4+4

6. A particle in one dimension is in an infinite potential well between

$$-\frac{L}{2} \leq x \leq \frac{L}{2}.$$

Consider a perturbation of the form

$$H' = \epsilon \cos\left(\frac{\pi x}{L}\right),$$

where ϵ is a small constant. Find the change in the energy of the ground state, to first order in ϵ .

5

7. Suppose you are given a quantum system whose Hamiltonian H_0 admits just two eigenstates, ψ_a with energy E_a , and ψ_b with energy E_b . They are orthogonal, normalized and non-degenerate (assume E_a is the smaller of the two energies). Now we turn on a perturbation H' , with the following matrix elements:

$$\langle \psi_a | H' | \psi_a \rangle = \langle \psi_b | H' | \psi_b \rangle = 0; \quad \langle \psi_a | H' | \psi_b \rangle = \langle \psi_b | H' | \psi_a \rangle = h,$$

where h is some specified constant.

- (a) Find the exact eigenvalues of the perturbed Hamiltonian.
 (b) Estimate the energies of the perturbed system using second order perturbation theory.
 (c) Estimate the ground state energy of the perturbed system using the variational principle, with a normalized trial function of the form

$$\psi = (\cos \phi) \psi_a + (\sin \phi) \psi_b,$$

where ϕ is an adjustable parameter.

2+2+2

8. The unperturbed Hamiltonian of a two-state system is represented by

$$H_0 = \begin{pmatrix} E_1^0 & 0 \\ 0 & E_2^0 \end{pmatrix}.$$

There is, in addition, a time-dependent perturbation

$$V(t) = \begin{pmatrix} 0 & \lambda \cos \omega t \\ \lambda \cos \omega t & 0 \end{pmatrix}, \quad (\lambda \text{ real}).$$

At $t = 0$, the system is known to be in the first state, represented by

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

Using time-dependent perturbation theory and assuming that $E_1^0 - E_2^0$ is not close to $\pm \hbar\omega$, derive an expression for the probability that the system is found in the second state, represented by

$$\begin{pmatrix} 0 \\ 1 \end{pmatrix},$$

as a function of t ($t > 0$).

4

9. The Hamiltonian for a one-dimensional harmonic oscillator in a laser electromagnetic field is given by

$$H = \frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega_0^2\hat{x}^2 + \frac{e\hat{p}}{2m\omega}E_0 \sin(\omega t) - \frac{1}{2}eE_0\hat{x} \cos(\omega t),$$

where ω_0 , m and e are the angular frequency, mass, and charge of the oscillator, and ω is the angular frequency of the radiation.

Assume the laser is turned on at $t = 0$ with the oscillator in its ground state ψ_0 .

- Write down the expression for $c_n^{(1)}(t)$ obtained in first order time dependent perturbation theory, necessary for the calculation of transition probability.
- Treat the electromagnetic interaction as a perturbation and find the transition probability for any time ($t > 0$) that the oscillator will be found in the first excited state.
 (You may use $\hat{x} = \sqrt{\frac{\hbar}{2m\omega}}(\hat{a}^\dagger + \hat{a})$; $\hat{p} = i\sqrt{\frac{m\hbar\omega}{2}}(\hat{a}^\dagger - \hat{a})$)

1+7



Indian Association for the Cultivation of Science
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Master's/Integrated Master's-PhD Program/Integrated Bachelor's-Master's Program/PhD Course
Mid-Semester Examination-Autumn 2025

Subject: Classical Mechanics

Subject Code(s): PHS 4101/AIS 4110

Full Marks: 25

Time Allotted: 2 h

Answer Question No. 1 (Q 1) and Any One from the Remaining Two

(Q1) (a) Let $y = y(x)$ denote a smooth function of x . Consider a given analytical function $f(y, y', x)$ where y' denotes the derivative with respect to x (i.e., f is a function of the path $y(x)$). Determine the path (i.e., an equation that leads to the path) for which the integral

$$J = \int_{x_1}^{x_2} f(y, y', x) dx$$

has a stationary value (here x_1, x_2 are two given points).

(6)

(b) (i) Explain the method of Lagrange's multiplier for optimizing a multivariable function in the presence of a holonomic constraint.

(6)

(ii) Find the extrema of the function $f(x, y, z) = x^2 + y^2 + z^2$, subject to the constraints $x + y + z = 1$ and $x^2 + y^2 = z$ using the method of Lagrange's multiplier.

(5)

(Q2) (a) Show that a Lagrangian is a scalar under a point transformation (a point transformation is a coordinate transformation where the new generalized coordinates are functions of the old generalized coordinates and time only, and not the old generalized velocities).

(4)

(b) A Newtonian system is described by an explicitly time-independent Lagrangian $L(q, p)$. Starting at (q_0, p_0) at $t = t_0$, the system is found at (q_f, p_f) at $t = t_f$. Now, another system is described by an explicitly time-dependent Lagrangian $L'(q, p) = L(q, p) + \kappa t \frac{dq}{dt} + \kappa q$, where κ is a constant. If the second system also starts from (q_0, p_0) at $t = t_0$, determine where it will be found at $t = t_f$.

(4)

(Q3) Consider a particle of mass m sliding from a higher point $P(x_i, y_i)$ to a lower point $Q(x_f, y_f)$ along a frictionless curve on a vertical (X-Y) plane containing those two points (take the X-axis to be horizontal, and the Y-axis to be vertically upward, and $y_i > y_f$). Considering uniform gravity acting vertically downward with the gravitational acceleration g .

(a) Express the time of fall of the particle from P to Q as a functional of the path $y = y(x)$ connecting P and Q. (4)

(b) Suppose the rate of change of a Lagrangian with respect to a particular generalized velocity (holding all other generalized velocities and the generalized coordinates fixed) is a constant κ . Determine the consequent symmetry transformation for the system. (4)



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**Master's/Integrated Master's-PhD Program/Integrated Bachelor's-Master's
Program/PhD Course**
End-Semester Examination-Autumn 2025

Subject: Classical Mechanics

Subject Code(s): PHS 4101/AIS 4110

Full Marks: 50

Time Allotted: 3 h

Answer question Q (1) [26 marks], and any 2 from the rest [12 + 12]

Q (1) (a) What are non-holonomic constraints? Give an example.

(2+2)

(b) A conservative system consists of N degrees of freedom which interacts via a potential that depends on the generalized coordinates only. The system is subjected to M independent holonomic constraints. Deduce the Euler-Lagrange equation for the system using the method of Lagrange's multiplier.

(6)

(c) Deduce Hamilton's canonical equations of motion for a system whose dynamics satisfies Euler-Lagrange equation of motion.

(5)

(d) Define canonical transformation and show that the following transformation is not canonical:

$$P = p \cos(q)$$

$$Q = p \sin(q)$$

(6)

(e) Define symmetry of a system executing dynamics described by a Euler-Lagrange equation of motion. Explain (with the relevant deduction) the circumstances under which a transformation can be a symmetry even when it changes the Lagrangian.

(2+3)

Q (2) (a) Define Poisson bracket $\{A, B\}$ between two observables A and B , and prove Poisson's Theorem which states that if $A(\mathbf{q}, \mathbf{p})$ and $B(\mathbf{q}, \mathbf{p})$ are two explicitly time independent observables, each of which is conserved under the relevant Hamiltonian dynamics of $\mathbf{p}$ and $\mathbf{q}$, then $\{A, B\}$ is also a conserved quantity, where $\mathbf{q}$ and $\mathbf{p}$ denote the sets of all canonical coordinates and momenta respectively. You can use the fact that Poisson bracket is a Lie bracket.

(1+4)

(b) Explain why the recursive use of the above process of generating conserved quantities cannot give an infinite number of conservation laws for a system with finite number of degrees of freedom.

↓
independent

(4)

(c) Deduce the time-evolution equation for an observable $A(\mathbf{q}, \mathbf{p})$ in terms of its Poisson bracket with the Hamiltonian. (3)

Q (3) (a) Prove Noether's theorem connecting continuous global symmetries of a Lagrangian with conservation laws. (4)

(b) A particle is moving in three dimensions in a potential which is given in the spherical polar coordinates (r, θ, ϕ) as $V = A \cos(\theta - \phi)/r$. Find the Noether charge (conserved quantity) corresponding to a continuous symmetry of the problem. (5)

(c) Using Noether's theorem deduce the Noether charge corresponding a generalized coordinate that is not explicitly present in the Lagrangian. (3)

Q (4) (a) Deduce the condition under which a Hamiltonian of a systems of Newtonian particles is the sum of the kinetic energy and the potential energy of the particles. (4)

(b) Let there be a quantity

$$\Gamma(\mathbf{q}, \mathbf{p}, t) = L - \sum_i p_i \dot{q}_i - \frac{d\beta(\mathbf{q}, t)}{dt},$$

where L denoting the Lagrangian of a system, $\mathbf{q}$ and $\mathbf{p}$ denoting the sets of all its canonical coordinates and momenta respectively, t denoting the time, and $\beta(\mathbf{q}, t)$ denoting an arbitrary analytic function of $\mathbf{q}$ and t only. Show that the total time-derivative of $\Gamma(\mathbf{q}, \mathbf{p}, t)$ vanishes if Γ has no *explicit* time dependence. (4)

(c) Deduce Hamilton's equations of motion by directly applying Hamilton's variational principle in the phase space. (4)

Q (5) (a) Two small beads of negligible dimensions of mass m each, are connected to each other via a spring of unstretched length l and spring-constant k . The beads are allowed to move only along the length of the spring. Setup and solve the small oscillation problem for the system and find the number of zero modes. (6 + 2)

(b) Suppose, the system is now placed freely on a frictionless plane. Assuming that it does not leave the plain but can move freely on it, explain how many zero modes the system will now have. (4)

Q (6) (a) Derive the condition on the Jacobian of a transformation for the transformation to be a canonical transformation. (4)

(b) Show that the set of all the matrices that are symplectic with respect to a fixed non-singular skew-symmetric matrix, form a group under matrix multiplication.

(4)

(c) If G is the generator of an infinitesimal canonical symmetry transformation, then G itself is the Noether charge corresponding to the continuous symmetry.

(4)

Best of Luck!



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Final-Semester (PG-I) Examination- November 2025

Subject: Mathematical Methods-I

Subject Code: PHS 4103/ AIS 4112

Full marks: 50

Time allotted: 3 hrs

Questions(Choose any 5 of the following 7 questions):

1. (a) The generating function of Hermite polynomial is defined by

$$e^{2xt-t^2} = \sum_{n=0}^{\infty} \frac{1}{n!} H_n(x) t^n$$

Evaluating the following integral

$$\int_{-\infty}^{\infty} e^{-x^2} e^{2sx-s^2} e^{2tx-t^2} dx$$

and comparing the coefficient of $s^n t^m$ show that

$$\int_{-\infty}^{\infty} H_n(x) H_m(x) e^{-x^2} dx = 2^n n! \sqrt{\pi} \delta_{nm}$$

[5]

[Hint: Use $\int_{-\infty}^{\infty} dx e^{-a(x+b)^2} = \sqrt{\frac{\pi}{a}}$]

- (b) The normalized Hermite function $\phi_n(x)$ is defined as

$$\phi_n(x) = \frac{1}{\sqrt{2^n n! \sqrt{\pi}}} H_n(x) e^{-\frac{x^2}{2}}$$

Show that

$$\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} ds \phi_n(s) e^{isx} = i^n \phi_n(x)$$

[5]

[Hint: Use the Fourier transform of the generating function $e^{2xt-t^2-\frac{x^2}{2}}$ and use the generating function of Hermite polynomial given above. The Fourier transform of e^{-ax^2} is $\sqrt{\frac{\pi}{a}} e^{-\frac{k^2}{4a}}$.]

2. (a) Find the Fourier transform of

$$f(x) = \begin{cases} 1 - x^2, & |x| < 1 \\ 0, & |x| > 1 \end{cases}$$

[5]

- (b) Using the inverse Fourier transform, show that

$$\int_0^\infty \left(\frac{x \cos x - \sin x}{x^3} \right) \cos \frac{x}{2} dx = -\frac{3\pi}{16}$$

[5] [Hint: No hints. Just be cool and perform standard high school integration and $e^{ikx} = \cos(kx) + i \sin(kx)$. At some point you may compare real part of the integration with 'something'.]

3. (a) Consider the modified Fourier series

$$f(\theta) = \sum_{n=-\infty}^{\infty} a_n r^{|n|} e^{in\theta}, \quad [0 < r < 1]$$

The coefficient

$$a_n \equiv \frac{1}{2\pi} \int_0^{2\pi} e^{-in\theta} f(\theta) d\theta$$

Show that

$$f(\theta) = \frac{1}{2\pi} \int_0^{2\pi} \left(\frac{1 - r^2}{1 - 2r \cos(\theta - \theta') + r^2} \right) f(\theta') d\theta'$$

[8] [Hint: Be careful about $|n|$. Use infinite GP sum formula $\sum_{n=0}^{\infty} br^n = \frac{b}{1-r}$.]

- (b) In which limit the Kernel defined in the above equation becomes a Dirac Delta function and why? [2]

4. In classroom, we have shown that a second order differential operator of the form

$$L = p_0 \frac{d^2}{dx^2} + p_1 \frac{d}{dx} + p_2,$$

can be written as Sturm-Liouville form

$$L = \frac{1}{\omega} \frac{d}{dx} \left(\omega p_0 \frac{d}{dx} \right) + p_2,$$

where $\omega(x)$ is a suitable weight function

$$\omega = \frac{1}{p_0} \exp \left[\int^x dx' \frac{p_1(x')}{p_0(x')} \right].$$

This is possible when p_0 is positive on the interval (a, b) and hence ω is also positive inside the interval.

(a) Show that,

$$\langle u, Lv \rangle_\omega - \langle Lu, v \rangle_\omega = [\omega p_0(u^* v' - u'^* v)]_a^b, \text{ where } \langle u, v \rangle_\omega = \int_a^b \omega u^* v dx$$

[5] *[Hint: We have done this in class!]*

(b) Reduce the following equations in the Sturm-Liouville form:

$$v'' - 2xv' + \lambda v = 0 \quad (1)$$

$$xv'' + (1-x)v' + \lambda v = 0 \quad (2)$$

[4]

(c) What will be the intervals (a, b) for those above equations from the requirement of positivity of ω ? [1]

5. (a) Consider the differential operator $L = \frac{d^4}{dx^4}$. Find the formal adjoint of L with respect to the inner product $\langle u, v \rangle = \int u^* v dx$ and find the corresponding boundary term $Q[u, v]$. [4]

(b) Find the adjoint boundary condition and the domain of $L^\dagger$ given the domain of L as

$$D[L] = \{y, y^{(4)} \in L^2[0, 1] : y(0) = y'''(0) = y(1) = y'''(1) = 0\}.$$

Show that $D[L] \neq D[L^\dagger]$. [3]

(c) What kind of boundary conditions makes the operator L to be truly self-adjoint? [3] *[Hint: Look at the surface term Q and guess it!]*

6. Find the solution of the boundary value problems by Green's function method:

$$(a) -y'' = f(x), y(0) = 0, y(1) + y'(1) = 0. \quad [5]$$

$$(b) -y'' - y = f(x), y'(0) = 0, y(1) = 0. \quad [5]$$

[Hint: Just follow the usual rules!]

7. (a) Find the Green's function for the following inhomogeneous ODE with the given boundary condition:

$$\left[x \frac{d^2}{dx^2} + \frac{d}{dx} - \frac{1}{x} \right] y = f(x), \text{ with } u(0) = u(1) = 0.$$

[7]

(b) Using the input of the above problem solve the following differential equation:

$$x^2 \frac{d^2 u}{dx^2} + x \frac{du}{dx} - (1 - \nu^2 x^2) u = 0, \text{ with } u(0) = u(1) = 0.$$

[3] *[Hint: Reduce the equation in the previous form.]*

Questions for Course PHS4111



1. Construct a class for matrix with n rows and m columns with float or double variables. Provide standard constructor as well as copy constructor. Provide methods for assignment, comparison, $+=$, $*=$ (with a scalar). Also provide multiplication of two matrices of sizes $L \times M$ and $M \times N$ and provide method of printing itself.
2. Construct a class `ThreeVector` with constructors with polar coordinates (r, θ, ϕ in float) as input: standard as well as default constructor (no input attributes); copy constructor; override methods for $=$, $==$, $+=$, $*=$ (with a scalar); providing methods for dot product, cross product. Also, provide methods to return each component and a method with ostream (to be used to print the content of three vectors).
3. Construct a generic class of Lorentz Vector (either working in real space or momentum space) which includes the three vector class to specify the space coordinates and has a method for Lorentz transformation with an external boost factor.
4. Generate random numbers in the range between 0 and 1 which follows the Gaussian distribution, $f(x) = \frac{1}{\sqrt{2\pi}} \exp\left[-\frac{x^2}{2}\right]$, with mean 0 and width 1 using the three standard methods: `memno` or transformation, the law of large numbers, and method of rejection.
5. An experiment measures the energy of a particle where the energy is shared among 9 detector elements: $E = \sum_i c_i H_i$. The coefficients are determined by exposing the detector to particles of fixed known energy so that the measured energy matches with the incident known energy and the spread in the energy measurement from 100 such measurements has the narrowest spread (best energy resolution).
6. The three angles of a triangle are measured each with its uncertainties. How the constraint relation be used to improve the measurements? Work out the likelihood method and



- Lorentz Transformation

$\vec{p}'(\vec{p}', \epsilon')$: Four momentum of B in rest frame of A

$P(\vec{P}, E)$: Four momentum of A in some frame

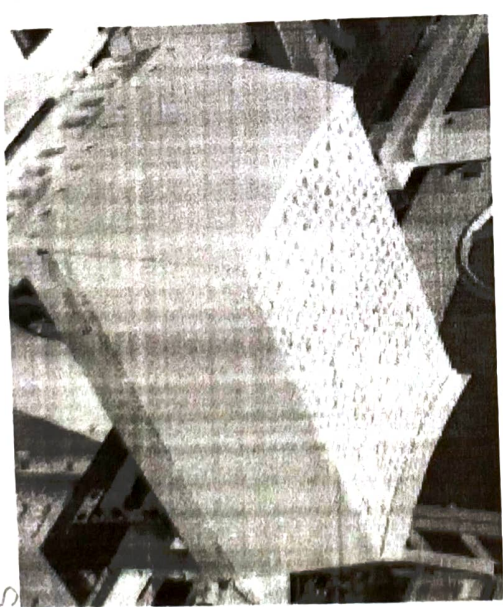
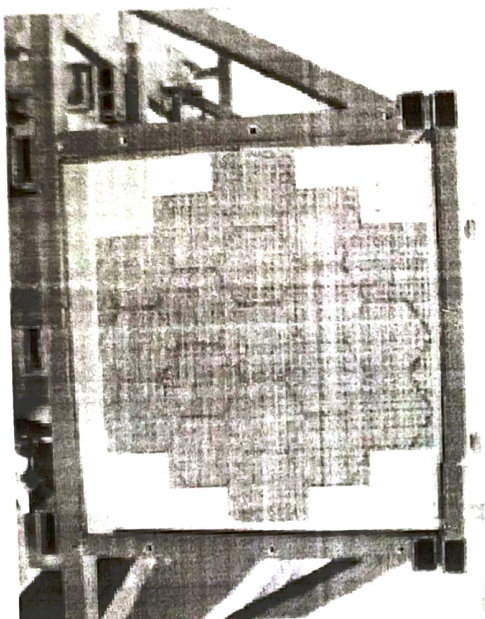
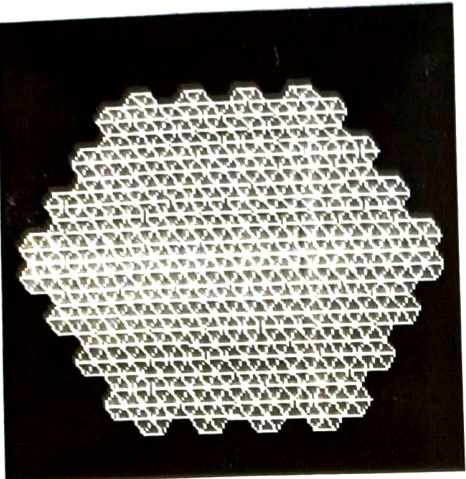
$p(\vec{p}, \epsilon)$: Four momentum of B in the same frame

$$\vec{p} = \vec{p}' + \vec{\beta} \gamma \left[\frac{\gamma}{\gamma + 1} \vec{\beta} \cdot \vec{p}' + \epsilon' \right]$$

$$\epsilon = \gamma [\epsilon' + \vec{\beta} \cdot \vec{p}']$$

where $\vec{\beta} = \frac{\vec{v}}{c}$ and $\gamma = \frac{1}{\sqrt{1 - \vec{\beta} \cdot \vec{\beta}}}$

- Calorimeter Cell: (some real-life illustrations)



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Master's/Integrated Master's-PhD Program/Integrated Bachelor's-Master's
Program/PhD Course
End-Semester Examination-Autumn 2025

Subject: Introduction to C++ & Numerical Methods

Full Marks: 50

Subject Code(s): PH4111

Time Allotted: 3 h

1. Construct a class for a matrix with n rows and m columns with float or double variables. Provide the standard constructor as well as the copy constructor. Provide methods for assignment, comparison, +=, *= (with a scalar). Also, provide the code for the multiplication of two matrices of sizes L x M and M x N and provide the code to be used for printing itself.
2. Construct a class ThreeVector with constructors with polar coordinates (r, θ , ϕ in float) as input: standard as well as default constructor (no input attributes); copy constructor; override methods for =, ==, +=, *= (with a scalar); providing methods for dot product, cross product. Also, provide methods to return each component and a method for ostream (to be used to print the content of three vectors).
3. Construct a generic class of Lorentz Vector (either working in real space or momentum space) which includes the three vector class to specify the space coordinates and has a method for Lorentz transformation with an external boost factor.
4. Generate random numbers in the range between 0 and 1, which follows the Gaussian distribution with mean 0 and width 1, using the three standard methods: method of transformation, the law of large numbers, and the method of rejection.
$$f(x) = \frac{1}{\sqrt{2\pi}} \exp\left[-\frac{x^2}{2}\right]$$
5. An experiment measures the energy of a particle where the energy is shared among 9 detector elements. The coefficients are determined by exposing the detector to particles of fixed known energy so that the measured energy matches the incident known energy, and the spread in the energy measurement from 100 such measurements has the narrowest spread (best energy resolution).
$$E = \sum_i C_i E_i$$
6. The three angles of a triangle are measured, each with its uncertainty. How the constraint relation can be used to improve the measurements? Work out the likelihood method and apply it to an example where the three angles are measured with uncertainties of 6 degree, 3 degree and 2 degree respectively.
7. Illustrate the terms: Random vs Systematic uncertainties, precision vs accuracy, and determinate vs indeterminate errors. In the example of finding the mass of resonance by fitting a distribution of the number of events as a function of energy, which could be random or systematic uncertainty of the measurement and how one can estimate them?

Lorentz Transformation:

$p' (\vec{p}', \epsilon')$: Four momentum of B in rest frame of A

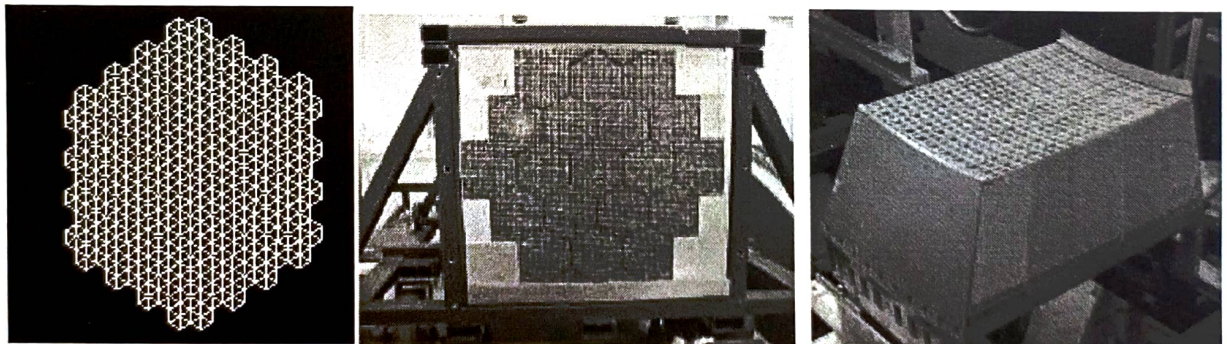
$P (\vec{P}, E)$: Four momentum of A in some frame

$p (\vec{p}, \epsilon)$: Four momentum of B in the same frame

$$\begin{aligned}\vec{p} &= \vec{p}' + \vec{\beta} \gamma \left[\frac{\gamma}{\gamma+1} \vec{\beta} \cdot \vec{p}' + \epsilon' \right] \\ \epsilon &= \gamma \left[\epsilon' + \vec{\beta} \cdot \vec{p}' \right]\end{aligned}$$

where $\vec{\beta} = \frac{\vec{P}}{E}$ and $\gamma = \frac{1}{\sqrt{1-\vec{\beta} \cdot \vec{\beta}}}$

Calorimeter Cell: (some real-life illustrations)





Indian Association for the Cultivation of Science
(Deemed to be University under *de novo* Category)

Master's/Integrated Master's-PhD Program/Integrated Bachelor's-Master's
Program/PhD Course
End-Semester Examination-Autumn 2025

Subject: Electronics
Full Marks: 50

Subject Code(s): PHS 4104/ AIS 4113/PHD 117
Time Allotted: 3 h

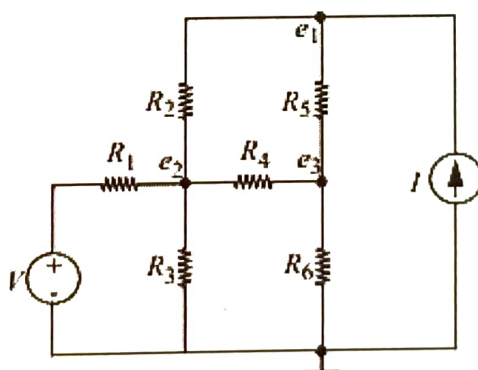
Circuit theory

1. A series RC circuit has $R = 1 \text{ k}\Omega$, $C = 1 \text{ }\mu\text{F}$, and initial capacitor voltage $v_C(0) = 0$. The input voltage is:

$$v_{\text{in}}(t) = 10(1 - e^{-1000t})$$

Find the Laplace transform $V_{\text{in}}(s)$. Also, derive the Laplace-domain capacitor voltage.

2. For the following circuit with unknown node voltages (e_1, e_2, e_3) , construct the matrix in the form $[G][e] = [S]$, where G is the 3×3 conductance matrix (by defining conductance $g_i = 1/R_i$) and S is the corresponding source vector. No need to solve for the node voltages.

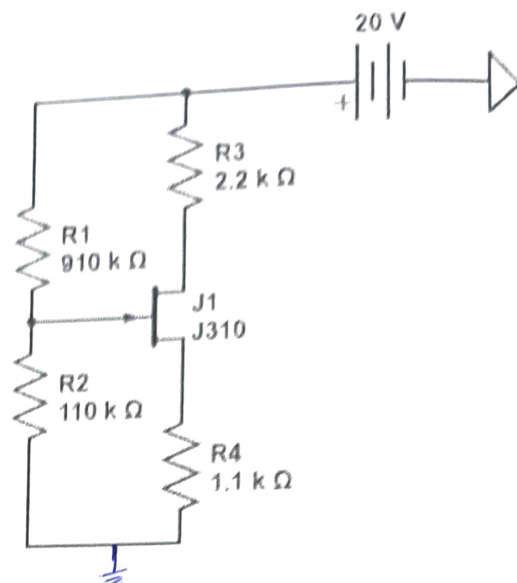


3. For the network with the n channel JFET, determine:

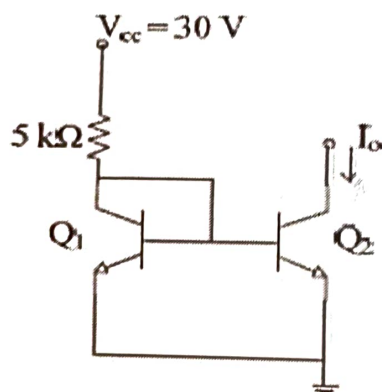
- a) V_G
- b) I_D and V_{GS}
- c) V_D and V_S
- d) I_{DS}

Given: $I_{DSS} = 10 \text{ mA}$

$V_P = -3.5 \text{ V}$



4. In the simple current source shown in the figure, Q1 and Q2 are identical transistors with current gain $\beta = 100$ and $V_{BE} = 0.7 \text{ V}$. Find I_0



5. A lock-in amplifier is used to measure a weak sinusoidal signal buried in noise. The reference signal is $v_{ref}(t) = 5 \cos(2\pi \cdot 1000t) \text{ V}$ and the input signal is $v_{in}(t) = 0.2 \cos(2\pi \cdot 1000t + \phi) \text{ V}$, where $\phi = 30^\circ$. The lock-in amplifier multiplies $v_{in}(t)$ by $v_{ref}(t)$ and applies a low-pass filter with a time constant large enough to average over many cycles. Calculate the DC output voltage (in mV) of the lock-in amplifier after low-pass filtering.

Semiconductor Device Physics

6. GaAs is a direct band gap semiconductor with $E_g = 1.42$ eV at 300 K. Take $N_c = N_v = 5 \times 10^{18} \text{ cm}^{-3}$ and independent of temperature.
- (a) Compute the intrinsic carrier concentration n_i .
 - (b) Demonstrate numerically how n_i can be increased by a factor of three *without introducing any dopants* and justify the physical basis of your approach.
- (2.5 + 2.5 marks)
7. Show schematically the junction formation between n-type semiconductor and metal where $\phi_m = \phi_s - 0.25$ eV where ϕ_m and ϕ_s are work functions for metal and n-type semiconductor, χ is electron affinity and E_g is band gap of n-type semiconductor.
- a. Determine whether the junction Schottky or Ohmic
 - b. Derive the expression for barrier height in equilibrium condition.
- (2.5 + 2.5 marks)
8. A p-i-n diode is formed using regions of the same semiconductor with the following doping levels:
- $N_A = 5 \times 10^{17} \text{ cm}^{-3}$, $N_D = 3 \times 10^{17} \text{ cm}^{-3}$, intrinsic layer thickness = $2.5 \mu\text{m}$.
- a. Draw a qualitative energy band diagram if the p-i-n diode in equilibrium, forward bias, and reverse bias.
 - b. Explain under which bias condition the diode delivers the maximum current.
- (2.5 + 2.5 marks)
9. A tunnel diode consists of a heavily doped p-n junction with $N_A = N_D = 2 \times 10^{19} \text{ cm}^{-3}$.
- a. Draw the band diagrams under: equilibrium and forward biases corresponding to: pre-peak region, peak current point, valley region, high forward bias
 - b. Using a plotted I-V curve, explain the origin of negative differential resistance (NDR)
- (2.5 + 2.5 marks)
10. A solar cell at room temperature is illuminated at $I_{\text{in}} = 600 \text{ W m}^{-2}$, producing: $I_{\text{SC}} = 180 \text{ mA}$, $V_{\text{OC}} = 0.55 \text{ V}$.
- a. Find the new values of I_{SC} and V_{OC} when the illumination is tripled.
 - b. Find the same quantities when the illumination is reduced to one-third. Explain the physical reasons behind the different scaling behavior of I_{SC} and V_{OC} .
- (2.5 + 2.5 marks)